9.3.4 The Rayleigh Quotient Iteration

9.3.4 Shifted Inverse Power Method, Part 1

fit width

A basic idea that allows one to accelerate the convergence of the inverse iteration is captured by the following exercise:

Homework 9.3.4.1. Let $A \in \mathbb{C}^{m \times m}$, $\rho \in \mathbb{C}$, and (λ, x) an eigenpair of A .

Which of the follow is an eigenpair of the ***shifted*** matrix $A - \rho I$:

- (λ, x).
- $(\lambda, A^{-1}x)$.
- $(\lambda - \rho, x)$.
- $(1/(\lambda - \rho), x)$.

▼ Answer ▼ Solution

Let $Ax = \lambda x$. Then

$$(A - \rho I)x = Ax - \rho x = \lambda x - \rho x = (\lambda - \rho)x.$$

We conclude that $(\lambda - \rho, x)$ is an eigenpair of $A - \rho I$.

$(\lambda - \rho, x)$.

Now justify your answer.

9.3.4 Shifted Inverse Power Method, Part 2

fit width

The matrix $A - \rho I$ is referred to as the matrix A that has been "shifted" by ρ. What the next lemma captures is that shifting A by ρ shifts the spectrum of A by ρ:

Lemma 9.3.4.1. *Let $A \in \mathbb{C}^{m \times m}$, $A = X \Lambda X^{-1}$ and $\rho \in \mathbb{C}$. Then $A - \rho I = X(\Lambda - \rho I)X^{-1}$.*

Homework 9.3.4.2. Prove Lemma 9.3.4.1 .

▼ Solution

$$A - \rho I = X \Lambda X^{-1} - \rho X X^{-1} = X(\Lambda - \rho I)X^{-1}.$$

This suggests the following (naive) iteration: Pick a value ρ close to λ_{m-1}. Iterate

$$\begin{array}{l} \mbox{Pick } v^{(0)} \mbox{ of unit length} \\ \mathbf{for}\ k = 0, \ldots \\ \quad v^{(k+1)} = (A - \rho I)^{-1} v^{(k)} \\ \quad v^{(k+1)} = (\lambda_{m-1} - \rho) v^{(k+1)} \\ \mathbf{endfor} \end{array}$$

Of course one would solve $(A - \rho I)v^{(k+1)} = v^{(k)}$ rather than computing and applying the inverse of A.

If we index the eigenvalues so that

$$|\lambda_{m-1} - \rho| < |\lambda_{m-2} - \rho| \leq \cdots \leq |\lambda_0 - \rho|$$

then

$$\|v^{(k+1)} - \psi_{m-1} x_{m-1}\|_{X^{-1}} \leq \left| \frac{\lambda_{m-1}-\rho}{\lambda_{m-2}-\rho} \right| \|v^{(k)} - \psi_{m-1} x_{m-1}\|_{X^{-1}}.$$

The closer to λ_{m-1} the shift ρ is chosen, the more favorable the ratio (constant) that dictates the linear convergence of this modified Inverse Power Method.

9.3.4 Rayleigh quotient iteration

fit width

A more practical algorithm is given by

$$\begin{array}{l} \text{Pick } v^{(0)} \text{ of unit length} \\ \textbf{for } k = 0, \ldots \\ \quad v^{(k+1)} = (A - \rho I)^{-1} v^{(k)} \\ \quad v^{(k+1)} = v^{(k+1)} / \|v^{(k+1)}\| \\ \textbf{endfor} \end{array}$$

where instead of multiplying by the inverse one would want to solve the linear system $(A - \rho I) v^{(k+1)} = v^{(k)}$ instead.

The question now becomes how to chose ρ so that it is a good guess for λ_{m-1}. Often an application inherently supplies a reasonable approximation for the smallest eigenvalue or an eigenvalue of particular interest. Alternatively, we know that eventually $v^{(k)}$ becomes a good approximation for x_{m-1} and therefore the Rayleigh quotient gives us a way to find a good approximation for λ_{m-1}. This suggests the (naive) Rayleigh-quotient iteration:

$$\begin{array}{l} \text{Pick } v^{(0)} \text{ of unit length} \\ \textbf{for } k = 0, \ldots \\ \quad \rho_k = v^{(k)\,H} A v^{(k)} / (v^{(k)\,H} v^{(k)}) \\ \quad v^{(k+1)} = (A - \rho_k I)^{-1} v^{(k)} \\ \quad v^{(k+1)} = (\lambda_{m-1} - \rho_k) v^{(k+1)} \\ \textbf{endfor} \end{array}$$

Here λ_{m-1} is the eigenvalue to which the method eventually converges.

$$\|v^{(k+1)} - \psi_{m-1} x_{m-1}\|_{X^{-1}} \leq \left| \frac{\lambda_{m-1}-\rho_k}{\lambda_{m-2}-\rho_k} \right| \|v^{(k)} - \psi_{m-1} x_{m-1}\|_{X^{-1}}$$

with

$$\lim_{k \to \infty} (\lambda_{m-1} - \rho_k) = 0$$

which means superlinear convergence is observed. In fact, it can be shown that once k is large enough

$$\|v^{(k+1)} - \psi_{m-1} x_{m-1}\|_{X^{-1}} \leq C \|v^{(k)} - \psi_{m-1} x_{m-1}\|_{X^{-1}}^2,$$

thus achieving quadratic convergence. Roughly speaking this means that every iteration doubles the number of correct digits in the current approximation. To prove this, one shows that $|\lambda_{m-1} - \rho_k| \leq K \|v^{(k)} - \psi_{m-1} x_{m-1}\|_{X^{-1}}$ for some constant K. Details go beyond this discussion.

Better yet, it can be shown that if A is Hermitian, then (once k is large enough)

$$\|v^{(k+1)} - \psi_{m-1} x_{m-1}\| \leq C \|v^{(k)} - \psi_{m-1} x_{m-1}\|^3$$

for some constant C and hence the naive Rayleigh Quotient Iteration achieves cubic convergence for Hermitian matrices. Here our norm $\| \cdot \|_{X^{-1}}$ becomes any p-norm since the Spectral Decomposition Theorem tells us that for Hermitian matrices X can be taken to equal a unitary matrix. Roughly speaking this means that every iteration triples the number of correct digits in the current approximation. This is mind-boggling fast convergence!

A practical Rayleigh Quotient Iteration is given by

$$\begin{array}{l} v^{(0)} = v^{(0)} / \|v^{(0)}\|_2 \\ \textbf{for } k = 0, \ldots \\ \quad \rho_k = v^{(k)\,H} A v^{(k)} \quad \text{(Now } \|v^{(k)}\|_2 = 1\text{)} \\ \quad v^{(k+1)} = (A - \rho_k I)^{-1} v^{(k)} \\ \quad v^{(k+1)} = v^{(k+1)} / \|v^{(k+1)}\| \\ \textbf{endfor} \end{array}$$

Remark 9.3.4.2. A concern with the (Shifted) Inverse Power Method and Rayleigh Quotient Iteration is that the matrix with which one solves is likely nearly singular. It turns out that this actually helps: the error that is amplified most is in the direction of the eigenvector associated with the smallest eigenvalue (after shifting, if appropriate).